4 Planning

You are to plan a series of experiments to verify that the order with respect to iodine in the iodination of propanone is zero.

The iodination of propanone is catalysed by acid, to form iodopropanone, as shown in the equation below.

Reaction 1: $CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I\ (aq) + HI\ (aq)$

You may assume that you are provided with:

- propanone, CH_3COCH_3
- dilute sulfuric acid, H_2SO_4
- aqueous solution of iodine, I_2
- sodium thiosulfate, $Na_2S_2O_3$
- sodium hydrogencarbonate, $NaHCO_3$
- starch indicator
- stopwatch
- the equipment normally found in a school or college laboratory

The order with respect to $[I_2]$ may be confirmed by preparing a reaction mixture containing H_2SO_4, I_2 and a large excess of CH_3COCH_3.

Portions of the reaction mixture are

- removed at timed interval,
- quenched by adding to an excess of $NaHCO_3$,
- titrated against a standard solution of $S_2O_3^{2-}$.

4 **(a)** Plan a procedure to collect sufficient data to allow a graph of volume of $S_2O_3^{2-}$ against time to be drawn.

You should plan to make a reaction mixture containing

- 25.0 cm^3 of aqueous propanone, CH_3COCH_3,
- 25.0 cm^3 of dilute sulfuric acid, H_2SO_4,
- 40.0 cm^3 of aqueous iodine, I_2.

This reaction mixture contains a large excess of propanone.

In your plan you should include brief details of:

- apparatus you would use,
- the procedure you would follow,
- the measurements you would make.

In your plan it is **not** necessary to refer to concentrations or to perform calculations.

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For Examiner's Use

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4 **(b)** Sketch on Fig. 4.1 the graph you would expect to obtain from carrying out your plan. Explain how you would verify if the order of reaction with respect to iodine is zero.

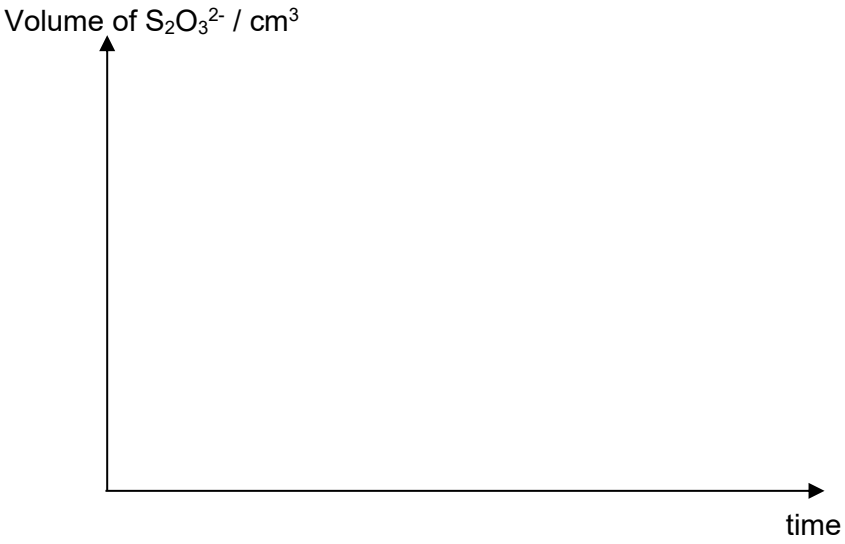

Fig. 4.1

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..... **[1]** M52

(c) Explain why the concentration of iodine used is very much lower than the concentration of propanone.

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Use

4 **(d)** Suggest **two** reasons why aqueous $NaHCO_3$ is preferred over aqueous NaOH as the reagent for quenching the reaction mixture.

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[Total: 10]